



Dementia

Mark Bosma, MD, FRCPC
Leah Nemiroff, MD, FRCPC

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Introduction

Dementia is a syndrome of acquired cognitive impairment that interferes with normal activities and function. The most common causes are Alzheimer disease (AD), vascular dementia (VD), a mixture of the two, Lewy body dementia (LBD) and frontotemporal dementia (FTD) (see [Table 1](#)). Dementia is also recognized as a complication of Parkinson disease (PD). The *Diagnostic and Statistical Manual of Mental Disorders Fifth Edition, Text Revision* (DSM-5-TR) has modified the nomenclature of the group of conditions formerly referred to as “dementias” to include other conditions that affect cognition, e.g., [Delirium](#) and traumatic brain injury; this new class of conditions is called neurocognitive disorders (NCDs).^[1] This chapter continues to use the traditional term of dementia, as most clinicians still utilize this terminology.

Dementias are almost always progressive, deteriorating illnesses (see [Table 2](#)). As dementia progresses, responsive behaviours may develop. Also known as behavioural and psychological symptoms of dementia (BPSD), they include depression, anxiety, apathy, agitation, delusions and hallucinations;^[2] see [Responsive Behaviours in Dementia](#) for more information.

Table 1: Common Dementia Types and Their Presentation

Type of Dementia	Pathophysiology	Hallmark Symptoms
Alzheimer ^[3]	Accumulation of beta-amyloid and tau proteins leads to intracellular plaques and extracellular tau tangles, both causing neuronal degeneration.	Gradual onset and progressive loss of cognition and memory, difficulty recalling recent events, sleep disturbances, apathy, apraxia.

Type of Dementia	Pathophysiology	Hallmark Symptoms
Vascular ^{[4][5]}	Cerebrovascular disease (e.g., multiple or single infarcts; cortical and/or subcortical involvement; ischemic, hemorrhagic or hypoperfusion mechanisms).	Stepwise decline in cognition after cerebrovascular events with cognitive deficits involving memory, executive function, language and attention.
Lewy body/Parkinson ^[6]	Aggregates of alpha synuclein proteins (Lewy bodies) in the brain stem, limbic and/or neocortical regions lead to neuronal loss. Amyloid plaques and neurofibrillary tangles may also be present.	Early-onset visual hallucinations, daytime drowsiness/napping, fluctuating consciousness, disorganized speech, parkinsonian symptoms, REM behavioural sleep disorder, antipsychotic sensitivity.
Frontotemporal ^[7]	Accumulation of abnormal deposits of intracellular proteins (including tau) leads to selective neurodegeneration of the frontal and temporal lobes. Specific proteins are associated with distinct molecular defects (e.g., FTLT-tau, FTLT-TDP-43, FTLT-FET, FTLT-UPS).	Disinhibition, socially inappropriate behaviour, compulsive behaviours, apathy, lack of empathy, change in habits and beliefs. Subtypes include prominent language deficits.

Table 2: Stages of Dementia

Stage	Characteristics	Corresponding FAST Rating ^[8]
Preclinical	Subjective complaints accompanied by very mild objective cognitive decline; functioning is unimpaired. This stage has considerable overlap with normal aging and may or may not progress to dementia.	3
Mild	Impaired instrumental activities of daily living (IADL), e.g., driving, medication use, finances, cooking, shopping, use of telephone/electronic devices and housekeeping.	4
Moderate	In addition to impaired IADL, some personal activities of daily living (ADL) such as bathing, grooming, dressing, toileting and feeding require assistance or prompting.	5

Stage	Characteristics	Corresponding FAST Rating ^[a]
Severe	Personal ADL cannot be done independently, even with prompting.	6
Terminal	Patients must be fed and become immobile and non-verbal.	7

^[a] Included because some jurisdictions use the [Functional Assessment Staging Tool \(FAST\)](#)^[8] in adjudicating reimbursement for dementia medications.

Goals of Therapy

- Alter the natural disease progression to optimize function and to meet patient and caregiver goals
- Manage cognitive, behavioural and psychological symptoms
- Mitigate risk of harm to self and others
- Minimize medication side effects
- Alleviate caregiver burden

Investigations

New concerns of cognitive impairment or otherwise unexplained decline in ability to perform instrumental activities of daily living require a careful history, including history from a collateral source such as a caregiver, to identify potentially reversible causes and timing of onset and progression. A detailed physical exam including a neurological exam, laboratory tests and neuroimaging should be conducted, if indicated (see [Table 3](#) and [Table 4](#)).

Table 3: Evaluation of Cognitive Impairment

Cognitive testing (screening)

- [Montreal Cognitive Assessment \(MoCA\)](#) is most useful for screening for mild cognitive impairment (MCI) or early dementia^[9]
- [Mini-Mental Status Exam \(MMSE\)](#), alone or in combination with the clock drawing test is sensitive and specific for distinguishing moderate stage dementia from normal cognition^{[10][11]}
- [Ascertain Dementia-8 \(AD-8\)](#), is completed by the caregiver^[11]
- [Dépistage Cognitif de Québec \(DCQ\)](#), includes the behavioural domain and has been validated for assessment of atypical dementias^[12]

- Brief screening tools may have test-specific biases and should be interpreted through an individualized lens. There is an increased risk of false positive screening and misclassification of degree of cognitive impairment for those of advanced age, nursing home residence, lower educational level and racialized backgrounds.^[13] Some tests like the MoCA have translated and culturally adapted versions, although the quality is variable^[14]

Patient history

- Cognitive history: duration, speed of onset and progression, domains of cognition affected (memory, language, orientation, visuospatial, executive function)
- Mood and behaviour history: personality changes, delusions, hallucinations, anxiety, depression, sleep, challenging behaviours
 - the neuropsychiatric inventory questionnaire (NPI-Q) can be used to help assess neuropsychiatric symptoms^[15]
 - the patient health questionnaire (PHQ) is helpful for screening mood symptoms^[15]
- Noncognitive symptoms: incontinence, gait disorder, dysphagia, constipation, parkinsonism
- Impact on function: functional disability is measured with tools such as the [Disability Assessment for Dementia \(DAD\)](#)^[16] or the [Functional Assessment Staging Tool \(FAST\)](#)^[8]
- Safety concerns (wandering, burning food)
- Social history

Medication review

Anticholinergic side effects of medications can lead to cognitive impairment.^[17] Having numerous prescribers, a frequent occurrence in older patients, is a risk factor for polypharmacy with anticholinergic agents.^[18]

- Explicit lists (e.g., BEERs^[19], STOPP/START^[20], Anticholinergic Cognitive Burden Scale^[21] and [calculator](#))
- Implicit guides (e.g., Medication Appropriateness Index)
- For the most commonly suspected medications, see [Table 4](#)

Physical exam

- Full physical exam including neurological exam^[22] can help identify the cause (e.g., parkinsonism, apraxia)
- Assess for potentially reversible medical illness that may be contributing to impairment (e.g., hypothyroidism, vitamin B₁₂ deficiency)

Laboratory tests

- CBC, fasting glucose, electrolytes, calcium, renal function, TSH, B₁₂^[23]

- Atypical presentations require additional specialized investigations that are best guided by a dementia expert (e.g., FDG-PET, SPECT, CSF) [\[15\]](#)
- Not recommended outside of research settings: biomarkers (e.g., CSF/plasma amyloid beta and tau) to assess for prodromal dementia (or “very early Alzheimer disease without dementia”) [\[24\]\[25\]\[26\]](#)

Neuroimaging

Consider neuroimaging (CT or MRI) broadly for most older adults with cognitive impairment. MRI is preferred if logistically feasible. Characteristics to help prioritize imaging if resources are limited include:

<60 years of age

New onset (less than 2 years of symptoms)

Rapid progression (1–2 months)

Unexpected and unexplained decline in cognition and/or function in a patient with known dementia

Recent significant head trauma or history of multiple mild head traumas

Use of anticoagulants or history of a bleeding disorder

Unexplained neurological symptoms including focal or lateralizing signs, gait disturbance

Early urinary incontinence and gait disorder (symptoms of normal pressure hydrocephalus)

Significant vascular risk factors

History of cancer, particularly if at risk of brain metastases

Unusual or atypical cognitive symptoms

Table 4: Medications That May Contribute to Dementia or Cognitive Impairment [\[17\]\[19\]\[27\]\[28\]](#)

Drug Class/Drug	Details
Note: higher total anticholinergic burden from combined medications with anticholinergic properties from any class may aggravate pre-existing dementia or contribute to potentially reversible cognitive impairment in someone without dementia; it is also associated with increased mortality.	
Antidepressants, e.g., amitriptyline, clomipramine, desipramine, doxepin, imipramine, nortriptyline, paroxetine [29]	Those with stronger anticholinergic properties may increase the risk of dementia. Nonanticholinergic antidepressants do not appear to increase dementia risk (e.g., other SSRIs, SNRIs).
Antiemetics/antivertigo agents, e.g., dimenhydrinate, promethazine, scopolamine	Not a known risk factor for dementia, but can contribute to acute cognitive impairment or delirium.

Drug Class/Drug	Details
Antihistamines, e.g., diphenhydramine, hydroxyzine	Not a known risk factor for dementia, but can contribute to acute cognitive impairment or delirium.
Antimuscarinics, e.g., darifenacin, fesoterodine, oxybutynin, solifenacin, tolterodine	Strong anticholinergic properties appear to increase risk of dementia and can contribute to acute cognitive impairment or delirium.
Antipsychotics, e.g., chlorpromazine, clozapine, olanzapine	Not a known risk factor for dementia but can cause delirium or worsen cognitive impairment. All antipsychotics are associated with an increased risk of cerebrovascular events and mortality in older patients with dementia; they should be used only if patient is at risk of self-harm or harming others.
Hypnotics, e.g., benzodiazepines, zolpidem, zopiclone [30]	Benzodiazepines may increase risk of dementia. All sedative-hypnotics can contribute to acute cognitive impairment or delirium.

Therapeutic Choices

Treatment choices are a combination of nonpharmacologic and pharmacologic strategies and vary based on the type and stage of dementia. For an overview of the management of dementia, see [Figure 1](#).

Nonpharmacologic Choices

- Most older adults with dementia benefit from ongoing exercise, cognitive stimulation and social engagement.
- People with mild to moderate dementia may benefit from group cognitive stimulation therapy. [\[31\]](#)
- The mainstay of FTD management is a nonpharmacologic approach targeted at managing symptoms, with emphasis on family and caregiver support.
 - Specific subtypes of people with FTD with prominent language issues may also benefit from early involvement of speech and language therapists. [\[32\]](#)
- Advance care planning should be addressed early, including establishing advance health-care directives (a living will) and durable powers of attorney before disease progression hampers competence.

- Psychosocial and psychoeducational interventions should be offered to caregivers of people with dementia.^[33] As resources vary considerably between provinces, refer patients and/or caregivers to their local [Alzheimer Society](#) as a starting point.
- Counsel patients against driving once they have progressed beyond the initial stages of disease. Many provinces have mandatory reporting of people living with moderate-stage dementia, prominent visuospatial or executive deficits, or hallucinations and delusions. See [Resources](#) for more information.

Pharmacologic Choices

Medications used in the management of patients with dementia are listed in [Table 5](#). Treatment choices are highly individualized and vary depending on dementia type and severity (see [Figure 1](#)).

Prior to consideration of treatment options, it is important to define target symptoms with patients and caregivers. Effectiveness is often considered to be either improvement or no deterioration of target symptoms as measured at regular intervals.^{[34][35]} Reduction in repetitive questioning is a common treatment goal in patients with mild to moderate AD and usually corresponds to a generally positive treatment effect.^[34] Monitor treatment effects 2–4 weeks after initiating therapy or increasing dosage and then periodically thereafter based on feasibility.

Although the [Mini-Mental Status Exam \(MMSE\)](#) is now recognized as a poor measure of clinically meaningful treatment response in individual patients, it remains a required measure for adjudicating reimbursement for cholinesterase inhibitors in some Canadian provinces. In untreated patients with mild to moderate dementia, a decline of 2–4 points per year on the MMSE scale is expected; therefore, an annual decline of less than 2 points while on drug therapy would typically indicate a beneficial effect.

Cholinesterase Inhibitors

The **cholinesterase inhibitors donepezil, rivastigmine and galantamine** block the breakdown of acetylcholine, resulting in increased acetylcholine levels in the synaptic cleft and increased function of neural cells. They are the mainstays of treatment for cognitive and functional symptoms in AD, VD, mixed AD/VD, LBD and PDD.^[15] They are not recommended in FTD, as they are not effective and may cause agitation. They may also have a role in managing behavioural and psychological symptoms, especially in mild to moderate dementia (see [Cholinesterase Inhibitors and Memantine](#)).

Donepezil is the only cholinesterase inhibitor indicated for all disease severities of AD; it is also used in mixed AD/VD, LBD and PDD.^[36] Benefits were seen with the 5 mg/day dose and were slightly larger in the 10 mg/day dose.^{[37][38][39]} There did not appear to be any additional benefit with doses higher than 10 mg/day. An advantage of donepezil is its once-daily dosing.

Rivastigmine is indicated for mild-moderate AD and mild-moderate PDD.^[36] A double-blinded trial with oral rivastigmine (twice daily dosing) in people with LBD and PDD demonstrated efficacy as well.^{[40][41]} Rivastigmine is also available in a transdermal patch that is formulated to provide continuous drug delivery over a 24-hour period. The patch is to be changed daily, which may be problematic in a patient with dementia. Health Canada has issued a warning regarding accidental overdoses from failure to remove one

patch before applying the next.^[42] Trials involving the rivastigmine patch showed a modest benefit compared with placebo and fewer side effects compared with the oral formulation.^{[43][44]} The patch is more expensive than oral formulations, and coverage varies among the provinces and private drug plans.

Galantamine is indicated for mild-moderate AD. It has been shown to be effective in several studies,^{[35][45]} including one in patients with mixed AD/VD.^[46] A Canadian multicentre trial suggested that patients treated with galantamine were more likely to meet clinical treatment goals than those on placebo.^[35] Due to limited evidence, galantamine is an option for patients with mild to moderate LBD only if donepezil or rivastigmine are not tolerated. Galantamine is available as an extended-release formulation administered once daily.

Alzheimer Dementia

While clinically detectable, benefits of cholinesterase inhibitors for AD are typically small to moderate.^[47] The benefit may consist mainly of disease stabilization and small improvements in activities of daily living and clinician-rated global clinical state.^[48] Because of variability in study populations, heterogeneity in symptomatology and disease stage at time of diagnosis, and the subjective nature of clinical stabilization (outside of objective testing), the duration of efficacy is variable between studies. In practice, we usually advise patients and caregivers that outcomes are highly individual and vary from no benefit to stabilization for a year or more.

A recent large retrospective health record study looked at data spanning 10 years and demonstrated that up to 68% of people with AD responded to cholinesterase inhibitors with an initial period of cognitive stabilization (2–5 months) before continuing to decline at the pretreatment rate.^[48] A large longitudinal cohort study of over 11 000 AD patients found a modest but persistent reduction in cognitive decline over a 5 year follow-up period compared to those with dementia but no cholinesterase inhibitor.^[49]

Donepezil, rivastigmine and galantamine have distinct structures and modes of action. In the absence of blinded, controlled, head-to-head trials, all 3 agents seem to be of similar efficacy when used as monotherapy for AD, and thus an individualized approach based on patient characteristics is often used.^[50]

Lewy Body Dementia/Parkinson Disease Dementia

In addition to cognitive stabilization, people with LBD/PDD on a cholinesterase inhibitor may also experience a reduction in visual hallucinations and other behavioural symptoms.^[51] Clinical trial data for the use of rivastigmine in LBD demonstrate that small doses can sometimes be effective, and leave the potential for subsequent dose titration over time.^[40] A placebo-controlled trial found statistically significant improvement in cognition, function and behaviour in rivastigmine-treated patients with PDD.^[41] A systematic review found that, as a class, cholinesterase inhibitors are effective in treating cognitive impairment in Parkinson disease and do not affect risk of falls. Although in theory procholinergic medications (such as cholinesterase inhibitors) can increase parkinsonian features, the review highlighted only an increase in tremor.^[52]

Vascular Dementia

A large Cochrane meta-analysis found that donepezil may have a slight beneficial effect on cognition in vascular dementia, although it is unclear if the benefits are clinically meaningful.^[53] For the most part, cholinesterase inhibitors are used in people with VD only if they have a mixed component of AD, LBD or PDD.^[36] In practice, the older the individual with dementia (even post-stroke dementia), the more likely they will have concomitant AD, suggesting that a trial of cholinesterase inhibitor warrants consideration for these patients.

Safety of Cholinesterase Inhibitors

Treatment response and side effects can vary among individuals; patients who do not respond to one cholinesterase inhibitor might respond to another. Typically, the overall treatment response is that some symptoms will improve, some will remain stable and others will worsen. For each of the drugs, doses at the higher end of recommended ranges typically have better outcomes.

Common adverse effects include diarrhea, nausea, vomiting, anorexia and/or weight loss, vivid dreams, tremor, vertigo, and other cholinergic effects (e.g., rhinorrhea, increased urinary frequency);^[54] follow-up to monitor for side effects is advised.

The cardiac safety of cholinesterase inhibitors has been reviewed in several reports with conflicting results. A population-based cohort study identified an increased risk of bradycardia and need for cardiac pacing, while a second cohort study suggested that the risk of cardiac events overall is not increased in patients receiving cholinesterase inhibitors.^{[55][56]} Another registry study suggested that all cholinesterase inhibitors have comparable cardiac safety profiles;^[57] Health Canada suggests that they all pose a small risk of QT interval prolongation.^[58] In practice, caution is generally recommended in patients with sick sinus syndrome, profound bradycardia, prolonged QT interval or multiple conduction abnormalities.

Tips for Prescribing Cholinesterase Inhibitors

- Start at the lowest dose and increase incrementally every 4 weeks as tolerated. The findings of a retrospective study suggest that starting cholinesterase inhibitors at high doses (without a slow up-titration) was associated with a higher rate of serious events, emergency room visits and hospitalizations.^[59]
- Avoid treating the side effects of a cholinesterase inhibitor with a new medication (e.g., oxybutynin to treat urinary frequency from a cholinesterase inhibitor); instead, consider reducing the dose or switching the cholinesterase inhibitor once the side effects have fully resolved after discontinuation of the initial agent.^[60]
- Practice varies considerably regarding cardiovascular screening prior to initiation of a cholinesterase inhibitor. Consider a baseline electrocardiogram (EKG) if the patient is bradycardic, on multiple cardiac rate-limiting agents (e.g., beta-blockers, calcium channel blockers, amiodarone), or has had unexplained syncope or presyncope—this may warrant discussion with a specialist prior to initiation of a cholinesterase inhibitor.

- If there is a specific concern about a person's ability to tolerate side effects, consider choosing a cholinesterase inhibitor with a shorter half-life, allowing for a faster washout period in the case of intolerable adverse effects. Donepezil has the longest half-life of all of the cholinesterase inhibitors (70 hours), compared to galantamine (6–8 hours), rivastigmine patch (3 hours) and rivastigmine oral (1.5–2 hours).^[61]
- Do not use cholinesterase inhibitors in FTD or in patients with mild cognitive impairment (not dementia).

Tips for Deprescribing Cholinesterase Inhibitors^{[15][62][63][64]}

- Decisions about if and when to discontinue a cholinesterase inhibitor should be individualized and not based on a single measure such as MMSE score, duration of treatment or institutionalization.
- Consider the values, preferences and experiences of the patient and the patient's family/caregiver when deciding if a trial of deprescribing is appropriate.
- Consider discontinuation of a cholinesterase inhibitor in the following situations:
 - patient or caregivers decide to stop or nonadherence
 - no clinically meaningful benefit (cognitive, functional and/or behavioural)
 - intolerable adverse effects
 - clinically meaningful progression of dementia over the past 6 months unexplained by anything else
 - progression to severe stage or terminal illness
- Avoid discontinuation of cholinesterase inhibitors for those with active psychotic symptoms, agitation or aggression (unless worsened after initiation or dose change).
- Continue treatment with cholinesterase inhibitors in people who have had clinically meaningful reduction in neuropsychiatric symptoms while on them.
- Taper by slowly reducing the dose to half of the previous dose (or the next-lowest available dose formulation) every 4 weeks.
- Monitor for emergent symptoms, including responsive behaviours, a rapid decline in cognition and/or a decline in function.^[65]
- If emergent symptoms or a significant deterioration in the patient's condition occurs within 1–3 months of discontinuation, reinstate the cholinesterase inhibitor at a low dose and titrate up to the last effective dose.
- See also [Deprescribing](#).

N-methyl-D-aspartate Receptor Antagonists

The proposed mechanism of action of the N-methyl-D-aspartate (NMDA) receptor antagonist **memantine** is to block glutamate-induced neuronal excitotoxicity, a process that is implicated as a final common pathway in neuronal death.

A Cochrane review suggests that memantine may have a small clinical benefit on cognition, activities of daily living (ADL), and behaviour and mood in moderate to severe Alzheimer disease.^[66] Due to limited evidence, memantine monotherapy is recommended in moderate Alzheimer disease only for patients who

are intolerant or have a contraindication to cholinesterase inhibitors; it may also be used in severe disease or to augment the response already achieved with cholinesterase inhibitors in moderate to severe disease.^{[36][67][68][69]}

Memantine may improve global impression of change in patients with LBD but not PDD, and is not effective in FTD.

Historically, the cost-effectiveness of cholinesterase inhibitors and of memantine was a barrier to treatment; however, it has become less of an issue with the introduction of generic alternatives with lower costs as well as the emergence of improved efficacy data.^[70] A cost-effectiveness trial in patients with moderate to severe Alzheimer disease taking **donepezil** showed there was benefit in continuing donepezil or starting memantine (if donepezil had been discontinued), but no cost-related benefit in combining the 2 agents.^[71]

When considering deprescribing memantine, considerations similar to those of the cholinesterase inhibitors may be applied. See [Tips for Deprescribing Cholinesterase Inhibitors](#).

Other Therapies

Lecanemab, an amyloid beta-directed antibody administered as a twice monthly IV infusion, was approved by Health Canada in October 2025 for the treatment of early-stage Alzheimer disease with confirmed amyloid pathology.^[72] In a randomized controlled trial, patients on lecanemab showed less cognitive and functional decline than those on placebo (0.45 points on the 18-point Clinical Dementia Rating–Sum of Boxes) over a period of 18 months.^[73] Significant adverse effects include brain swelling and bleeding (also known as amyloid-related imaging abnormalities or ARIA) at an incidence of 17%; most cases were asymptomatic but detectable on PET/MRI scans. Lecanemab is not yet marketed in Canada; significant resources required for patient selection, administration, monitoring and reimbursement may impede public funding. No further information can be provided at this time.

Vitamin supplementation remains controversial. Only a single study of high-dose vitamin E (2000 units alpha-tocopherol per day) appeared to demonstrate slowing of dementia progression but more recent data suggests no benefit and potential risk of harm, so use of high-dose vitamin E is not recommended.^{[74][75]}^[76] A trial of high-dose vitamin B supplementation did not slow cognitive decline in people with mild to moderate Alzheimer disease.^[77]

Other medications including statins, anti-inflammatory drugs, stimulants and Ginkgo biloba cannot be recommended either for the treatment or prevention of Alzheimer disease due to lack of evidence.^[78]

Therapeutic Tips

- Tips for monitoring treatment effects on target symptoms:

- In certain provinces, reimbursement of drug costs by the provincial drug plan is predicated on monitoring degrees of improvement/worsening of target symptoms, such as repetitiveness, decreased initiative and impaired recent memory. To effectively monitor target symptoms, they should be described precisely, e.g., instead of “impaired function,” state “cannot carry on a conversation; will call only 1 or 2 well-known numbers; cannot remember how often he has called.” For many symptoms, it is important to quantify how often they occur, e.g., “asks the same question up to 20 times per day.”
- See [Tips for Prescribing Cholinesterase Inhibitors](#) and [Tips for Deprescribing Cholinesterase Inhibitors](#).

Responsive Behaviours in Dementia

Investigations

The responsive behaviours in dementia are defined as the behavioural and psychological manifestations of dementia, e.g., delusions, hallucinations, anxiety, depression, agitation and apathy.^[2] A strategy for evaluating behavioural problems in the older adult is illustrated in [Figure 2](#).

Responsive behaviours in dementia (RBD), also known as behavioural and psychological symptoms of dementia (BPSD) are seen in 3 contexts:

- Related to the *neurocognitive* disorder. These symptoms often occur predictably, according to the stage of dementia. In Alzheimer disease, for example, depression and apathy often occur early, when the dementia is mild. In contrast, psychosis and agitated behaviour tend to occur in the moderate to severe stages. Current research is assessing the “noncognitive” prodrome of dementia in which people with no or minimal cognitive impairment demonstrate mild behavioural impairment.
- Related to *psychiatric* illness that predates the dementia, such as depression, anxiety, psychosis, and personality issues. It is important to elicit a careful psychiatric history, as symptoms of psychiatric illness that occurred earlier in the patient’s life may recur (often in slightly different form, reflecting cognitive impairment) and can guide treatment.
- Related to a *delirium-like* presentation, typically associated with a medical and/or environmental precipitant^[79] (see [Figure 2](#)). The diagnostic approach in the setting of delirium includes:
 - physical examination to look for signs of an underlying medical cause, such as infection, cardiac failure or neurological syndrome
 - review of medications, including recent changes and prescribing of medications that can cause delirium
 - laboratory tests to look for an underlying medical cause, which may include CBC, electrolytes, urea, creatinine, glucose, urinalysis and chest x-ray
 - for more information, see [Delirium](#)

Note: It is unusual for delusions or hallucinations to occur in early dementia. When this happens, it can indicate that the person is suffering from Lewy body dementia. Fluctuation in the level of consciousness,

a hallmark of delirium, can also be seen in patients with Lewy body dementia. Increasingly, delirium that either persists or only partly resolves is being recognized;^[80] thus, there can be significant difficulty in distinguishing delirium from dementia, particularly Lewy body dementia.

Therapeutic Choices

Nonpharmacologic Choices

Nonpharmacologic approaches are first-line therapy for responsive behaviours due to their low risk of adverse effects. They involve psychosocial and environmental interventions and are often sufficient to make a noticeable improvement in the target symptoms.

Depression and anxiety symptoms in patients with mild dementia may be reduced via structured psychological interventions (including cognitive behavioural therapy [CBT] and a supportive counselling approach).^[81]

Agitation may be managed through multidisciplinary care (e.g., recreation therapy, exercise and outdoor activities) and sensory stimulation (e.g., massage and therapeutic touch, with or without music).^[82] Supervised skills training for home care staff is effective and involves education regarding person-centred care, communication and dementia care mapping. Interventions with insufficient or no evidence include light therapy, aromatherapy, and training family caregivers in behavioural management therapy (BMT) or CBT.^[83]

Environmental interventions and lifestyle modifications include modifying the physical environment to ensure safe places to wander and removing environmental hazards, as individuals with dementia are at increased risk of accidents such as falls and burns.

- If pharmacologic therapy is deemed necessary, concurrent nonpharmacologic therapies should always be used, as they often allow medications to be used more sparingly.^[84]
- All interventions should begin with a clear and respectful communication with the patient and caregivers.
- All caregivers, including family and multidisciplinary team members, should be involved in all nonpharmacologic therapy. Caregivers should be educated on strategies such as distraction, redirection, avoiding confrontation, reassurance, and providing stimulation and structure.

Pharmacologic Choices

Antidepressants

Many patients in the early stages of dementia experience depression. Occasionally, depression can manifest with prominent cognitive impairment in patients without dementia. Previously known as

pseudodementia, it is also referred to as dementia of depression. A significant proportion of these patients will progress to dementia.

Treatment of depression in patients with dementia remains challenging, given the lack of evidence for use of antidepressants in this situation. Although individual studies exist that suggest significant benefit, overall there is a lack of evidence for their efficacy, as demonstrated by a comprehensive Cochrane review.^{[85][86][87][88][89][90][91][92]} Guidelines do not recommend the routine use of antidepressants to manage mild to moderate depression in people living with mild to moderate dementia, unless they have a pre-existing mental health problem.^[36] However, many psychiatrists would still consider prescribing antidepressants for patients with dementia, taking into account several factors, including stage of dementia, severity of symptoms, and availability and feasibility of psychological treatments.

When choosing an antidepressant, it is important to take several factors into account. **SSRIs** (with the exception of **paroxetine**) are less likely than tricyclic antidepressants (**TCAs**) to cause anticholinergic side effects or to worsen orthostatic hypotension, which are common and problematic in this population. An increased risk of hyponatremia/SIADH (syndrome of inappropriate antidiuretic hormone secretion), possibly potentiated by concurrent thiazide use, has been reported in older patients taking SSRIs.^{[93][94]} Monitoring of electrolytes should be considered in this context because of the effect of hyponatremia on cognitive function. It is generally recommended to avoid using TCAs in dementia patients.^[32] If using a TCA is necessary because of lack of response or sensitivity to an SSRI, choose **desipramine** or **nortriptyline** because of their lower propensity to cause anticholinergic effects. Older adults with depression often require longer trials of antidepressant treatment, with signs of improvement often occurring after 6–8 weeks of treatment (see [Depression](#) for dosing information).

Limited evidence suggests that antidepressants improve nondepression-related RBD;^[95] antidepressants are commonly used for anxiety (including compulsive behaviour), agitation and sexually inappropriate behaviour. A Cochrane review reported that **sertraline** and **citalopram** reduced symptoms of agitation in dementia patients when compared with placebo in 2 studies.^[96] An RCT found that citalopram was as effective as risperidone in treating either agitation or psychotic symptoms in patients with dementia.^[97] Another study demonstrated that citalopram (up to 30 mg per day) significantly reduced agitation and caregiver distress; however, its use is often limited by potential cardiac effects (prolonged QT_c interval).^[98]

In contrast, a recent RCT suggests that mirtazapine is not appropriate for management of agitation in dementia, given its lack of efficacy and possible trend toward increased mortality.^[99]

Tips for Using Antidepressants in Dementia

- Start at a low dose.
- Monitor patient for side effects.
- Continue to increase the dose as side effects permit until the patient benefits or the maximum dose has been reached; effective doses will be similar to those used to treat younger adults.
- Consider maintaining treatment at a therapeutic dose for up to 6–8 weeks before evaluating success of the treatment.

Antipsychotics

Use of antipsychotic medication in treatment of RBD is challenging due to the significant risk of adverse effects. Evidence suggests that symptoms can improve with treatment, but which patients will respond cannot be determined in advance. The American Psychiatric Association recommends that antipsychotic medication be reserved for the treatment of agitation or psychosis in patients with dementia when symptoms are severe, dangerous or cause clinically significant distress.^[100]

Because older patients with dementia are very sensitive to antipsychotics, initial doses should be small.

Risperidone and **olanzapine** have the most evidence for the treatment of RBD, including a Cochrane review that demonstrated utility for reducing aggression, and as such are often the initial choice of treatment.^[101] **Quetiapine** and **aripiprazole** have limited evidence to support their use in dementia patients.^{[102][103]} First-generation antipsychotics are generally not recommended for the management of RBD; a Cochrane review of **haloperidol** for agitation in dementia recommended against its routine use.^[104]

Older patients are at greater risk of extrapyramidal side effects (EPS), which include parkinsonism, akathisia, dystonia, and tardive dyskinesia, a potentially irreversible movement disorder. EPS is most commonly seen with first-generation antipsychotics but can also occur with second-generation agents (especially with doses over 2 mg/day of risperidone). In patients over 60 years of age treated with a first-generation antipsychotic, tardive dyskinesia (TD) occurs in 29% of patients after 1 year and in up to 63% after 3 years of use.^[105] A 9-month study found a 2.6% incidence of TD with risperidone.^[106] For patients with baseline movement disorders or those who develop EPS with other second-generation antipsychotics, quetiapine may be the best option given its lower risk of EPS. The Clinical Antipsychotic Trials of Intervention Effectiveness (CATIE) study in Alzheimer disease showed that the chance of benefiting from a second-generation antipsychotic (risperidone, olanzapine, quetiapine) was equal to the risk of experiencing an adverse effect such as EPS, parkinsonism, sedation or confusion/mental status changes.^[107]

Patients with Lewy body dementia (a condition that involves deficits in dopamine) are at even greater risk of side effects from antipsychotic medication, due to their antidopaminergic properties. Antipsychotics may worsen psychosis in patients with Lewy body dementia^[108] and can lead to severe antipsychotic sensitivity reactions that can precipitate irreversible parkinsonism, further impair consciousness and induce autonomic disturbances. Treatment with an antipsychotic should be considered only when therapy with cholinesterase inhibitors has failed and responsive behaviors are causing severe distress. Limited evidence suggests that second-generation antipsychotics, specifically quetiapine and olanzapine, may be better tolerated than first-generation agents.^[109]

The risk of serious adverse events with all antipsychotics is significant in older patients with dementia. An increased risk of stroke and death has been demonstrated with second-generation antipsychotics compared with placebo, and observational comparisons suggest the risk may be even higher with first-generation antipsychotics.^{[95][110][111]} A retrospective, case-controlled study identified the number needed to harm (NNH; i.e., number of patients who receive treatment that would be associated with 1 negative outcome) in antipsychotic use versus nonuse and found that haloperidol was associated with a 3.8% relative mortality risk (NNH = 26) followed by risperidone (3.7%; NNH = 27), olanzapine (2.5%; NNH = 40) and quetiapine (2.0%; NNH = 50).^[112]

Health Canada has issued an advisory stating that the use of second-generation antipsychotics in older patients with dementia was associated with a 1.6 times greater mortality rate compared with placebo.^[113] It has also warned that there is a risk of cerebrovascular events with these medications, especially in patients with mixed and vascular dementia.^[114] Thus, the indication for risperidone in dementia has been restricted to the short-term management of aggression/psychotic symptoms in patients with severe Alzheimer dementia if there is risk of harm to self/others and nonpharmacologic treatment has failed.^[114]

Tips for Using Antipsychotics in Dementia

- Reserve for patients with severe responsive behaviours, particularly with aggression that poses risk of harm to self or others, if nonpharmacologic measures are ineffective.
- Use a second-generation agent first, with caregiver support and consent.
- Start low and go slow, and use the lowest possible effective dose.
- Select target symptoms and aim for improvement in symptoms rather than complete resolution.
- Monitor frequently for side effects, including EPS. Antipsychotic-induced akathisia (increased motor restlessness) may be misinterpreted as lack of drug effect. This can initiate a cycle of increased antipsychotic use and worsening akathisia, and result in other adverse effects occurring due to the escalating doses.
- Use quetiapine in patients with LBD/PDD if an antipsychotic agent is required.
- Evaluate the need for antipsychotic medication every 3–6 months, as symptoms may subside naturally and allow for tapering of the antipsychotic.
- [Deprescribing.org](#) offers an evidence-based algorithm to assist with deprescribing anti-psychotics.^[115]
- See also [Deprescribing](#).

Cholinesterase Inhibitors and Memantine

As mentioned above, cholinesterase inhibitors are the mainstay of treatment for cognitive and functional symptoms of dementia. They are also now commonly used for the treatment of RBD, with less risk of serious adverse events compared with antipsychotics. Multiple RCTs have shown statistically significant improvement in RBD with cholinesterase inhibitors; however, the overall effect on these behaviours appears to be small, suggesting questionable clinical significance.^{[95][116][117]} Evidence for the use of memantine is similar, with statistically significant improvement that may be undetectable by clinicians.^[116]
^[117]

Trazodone for Behavioural and Sleep Disturbances

Trazodone, a serotonergic antidepressant, is often used successfully to manage agitated behaviour, with some RCT evidence to support its use.^[118] Trazodone is also used to treat disrupted sleep/wake cycles and “sundowning” (worsening of behaviour as darkness falls). In a systematic review of various medications used to treat sleep disturbances associated with Alzheimer disease,^[119] low-dose trazodone had some evidence of a beneficial effect based on an RCT of 30 patients. In this small trazodone study, 50 mg QHS administered for 2 weeks to community-dwelling patients with AD increased daily sleep time by 43 minutes with no cognitive or functional impairment and no other serious adverse effects.^[120] The review also concluded that no benefit was seen with melatonin and ramelteon (not available in Canada); studies for other widely prescribed drugs (e.g., lorazepam, oxazepam, temazepam, triazolam, tryptophan, zolpidem) were not RCTs and were excluded from the meta-analysis.^[119]

Benzodiazepines

Data on the efficacy of **benzodiazepines** for RBD are conflicting. Although their use can result in oversedation, falls and worsening cognition, benzodiazepines are sometimes indicated for severe agitation, especially when other agents fail. Low doses of a short-acting agent without active metabolites (e.g., **lorazepam** 0.5–1 mg, **oxazepam** 5–10 mg, **temazepam** 15 mg) may be tried. In an acute situation, to manage severely agitated patients, lorazepam 0.5–1 mg can be mixed in the same syringe with **haloperidol** (0.5, 1 or 1.5 mg) and given IM every 8 hours for a maximum of 3 days. A double-blinded, randomized, controlled comparison of IM lorazepam and IM **olanzapine** found them to be equally effective.^[121]

Others

Beta-blockers (particularly **pindolol**), **carbamazepine**, **divalproex**, **lithium**, **buspirone**, **gabapentin** and **pregabalin** have been used successfully in case reports, but more robust evidence is lacking. These agents seem to work best when the problem behaviour mimics the psychiatric syndrome for which the drug is efficacious (e.g., lithium for manic features) or when medications with a stronger evidence base have failed.

Methylphenidate may be effective in the management of apathy in patients with Alzheimer disease, as well as having small benefits for cognition and functional performance; however, higher-quality evidence is required before adopting this practice.^[122]

Prevention of Dementia

Investigations

Routine screening for cognitive impairment in asymptomatic older adults has no evidence of benefit and is not recommended, even in the presence of risk factors such as vascular disease or family history.^[15] However, a growing body of work makes it clear that the risk of dementia and cognitive decline in later life is related in part to physical health.^{[123][124]}

It is estimated that up to 40% of dementia cases can be attributed to 12 potentially modifiable risk factors that impact cognitive reserve in early life and influence neuropathological developments in mid and later life.^[125] These risk factors include less education (to a maximum of 11–12 years of age), social isolation, late-life depression, mid-life obesity, physical inactivity, hearing impairment, diabetes, mid-life hypertension, smoking, air pollution, traumatic brain injury and alcohol consumption (>21 units/wk).

Therapeutic Choices

Nonpharmacologic Choices

Addressing the known potentially modifiable risk factors for dementia is an important step in the nonpharmacologic prevention of dementia for older adults. This includes social programs and supports to address social isolation,^[126] early identification and management of hearing impairment,^[127] alcohol and smoking risk-reduction strategies,^[125] and lifestyle modifications to address vascular risk factors and physical inactivity.^[128]

Physical exercise may be an effective intervention in improving cognitive outcomes in people with mild cognitive impairment and reducing the risk of dementia. A recent longitudinal observation study found that greater physical activity appeared to attenuate the negative association between amyloid-beta burden and cognitive decline in asymptomatic people. In order to confer consistent benefit for those with cognitive impairment, it appears that 30 minutes of exercise more vigorous than a brisk walk at least 3 times per week is the minimum amount required.^[129] A balance of moderate intensity aerobic and resistance training, along with dance and mind-body exercise (such as tai chi), may improve cognitive outcomes in older adults. although large high-quality studies are lacking.^[130] Regardless, there is likely no harm in making this recommendation to an older adult seeking dementia risk-reduction.

A variety of mentally stimulating activities in late life may lower the risk of cognitive impairment. Prospective cohort studies have demonstrated that engaging in activities such as playing games, computer use and social activities reduced the risk of developing mild cognitive impairment (MCI), often a precursor to dementia.^[131] As well, a longitudinal observational study found that participation in intellectual activities was associated with a lower risk of dementia independent of health-related and social factors.^[132]

Diet may affect cognitive decline and incidence of dementia. Current evidence suggests that following a Mediterranean diet with high levels of mono- and polyunsaturated fatty acids, low intake of saturated fatty acids, and high consumption of fruits and vegetables can be a protective factor against cognitive decline.^{[15][133]}

Identification and management of sleep apnea may also improve cognition and reduce risk of dementia.^[15] Avoiding severe sleep deprivation (<5 h/night) may improve cognition or reduce the risk of dementia. Multiple studies have shown an association between sleep disruption and cognitive impairment, but there is insufficient evidence of causality to determine if management of other sleep disorders (e.g., sleep fragmentation, circadian rhythm disruption) can reduce risk of dementia.

Pharmacologic Choices

Medication review and deprescribing of sedating medications and/or those with high anticholinergic burden can improve cognitive outcomes in older adults.^[134] When identified, dose reduction and/or rotation to alternate agents should be considered for all older adults to reduce cognitive risk (see [Table 4](#)).

Several clinical trials support the use of antihypertensive agents for dementia prevention.^{[135][136][137][138]} One recent meta-analysis reviewed prospective community-based studies with long-term follow-up of >31 000 participants and found that use of any antihypertensive agent reduced the risk of dementia in those with baseline hypertension.^[139]

Using **HMG-CoA reductase inhibitors (statins)** solely for the prevention of dementia remains controversial. Some systematic reviews have shown that statins may have a preventive role,^[140] especially in those at higher risk of dementia, such as carriers of the ApoE4 genotype (which predisposes individuals to a rise in amyloid-beta peptide);^[141] however, a Cochrane review remains more skeptical.^[142]

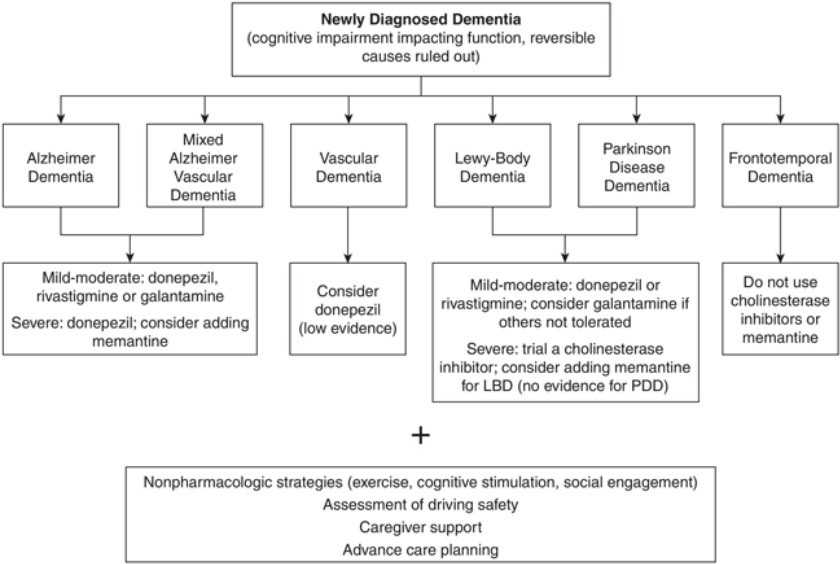
There does not appear to be a role for the use of NSAIDs^[143] or estrogen^[144] in reducing dementia risk.

The use of **vitamins** to prevent dementia is controversial. Vitamin E is not recommended for prevention of dementia;^[75] the PREADViSE trial, a large observational cohort study, failed to show benefit from vitamin E and selenium in the prevention of dementia.^[145] Lower vitamin D levels have been associated with an increased risk of dementia in a systematic review,^[146] but interpretation of these results can be difficult, as many studies were observational and lacked adjustment for confounding factors. Clinical trials in people with mild cognitive impairment demonstrate that **vitamin B** supplementation may be beneficial in preventing progression but only in those with elevated baseline levels of homocysteine.^{[147][148]}

Medications used for the treatment of dementia, such as cholinesterase inhibitors and memantine, are not effective for prevention of progression of mild cognitive impairment or preclinical dementia (see [Table 2](#)).^[78]

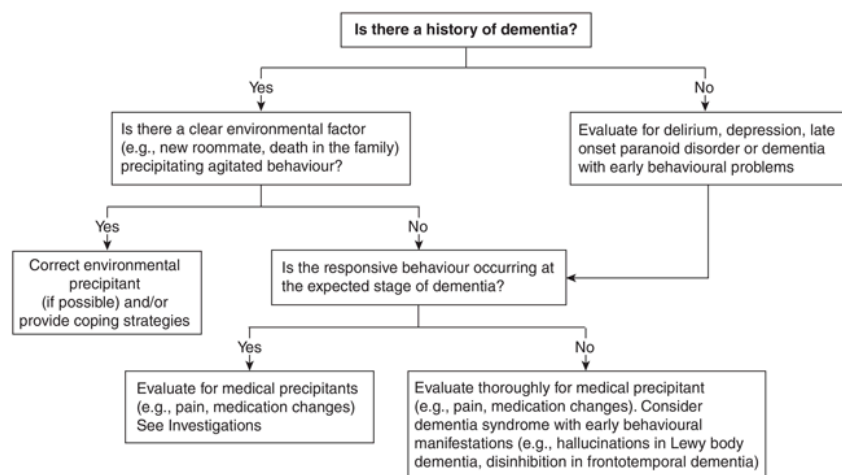
Algorithms

Figure 1: Management of Newly Diagnosed Dementia



LBD = Lewy-body dementia; PDD = Parkinson disease dementia

Figure 2: Evaluation of the Older Adult with Responsive Behaviours



Drug Table

Table 5: *Drugs Used for the Management of Dementia*

Drug/Cost ^[a]	Indications	Dosage	Adverse Effects	Drug Interactions
Drug Class: Antipsychotics				
olanzapine Zyprexa , Zyprexa Zydys, generics < \$30	• RBD : psychosis, agitation	Initial: 2.5 mg/day PO Usual: 5– 10 mg/day PO Maximum: 10 mg/day PO	Extrapyramidal symptoms (EPS), sedation, constipation, GI upset, weight gain, metabolic dysregulation (type 2 diabetes); increased risk of stroke and death.	Toxicity may be increased by inhibitors of CYP1A2 such as ciprofloxacin, fluvoxamine, ketoconazole. Effectiveness may be decreased by inducers of CYP1A2 such as carbamazepine, cigarette smoking,

Drug/Cost ^[a]	Indications	Dosage	Adverse Effects	Drug Interactions
phenobarbital, primidone, rifampin.				
quetiapine Seroquel, generics < \$30	• RBD : psychosis, agitation	Initial: 12.5 mg/day PO Usual: 25– 50 mg/day PO Maximum: 200 mg/day PO	Extrapyramidal symptoms (EPS), sedation, constipation, GI upset, weight gain, metabolic dysregulation (type 2 diabetes); increased risk of stroke and death.	Toxicity may be increased by inhibitors of CYP3A4 such as clarithromycin, erythromycin, grapefruit juice, ketoconazole. Effectiveness may be decreased by inducers of CYP3A4 such as carbamazepine, phenytoin, rifampin.

Drug/Cost ^[a]	Indications	Dosage	Adverse Effects	Drug Interactions
risperidone  Risperdal , generics < \$30	• RBD: psychosis, agitation	Initial: 0.25 mg/day PO Usual: 1 mg/day PO Maximum: 2 mg/day PO	Extrapyramidal symptoms (EPS), sedation, constipation, GI upset, weight gain, metabolic dysregulation (type 2 diabetes); increased risk of stroke and death.	Toxicity may be increased by inhibitors of CYP2D6 or CYP3A4 such as erythromycin, grapefruit juice, paroxetine, prednisone. Effectiveness may be decreased by inducers of CYP2D6 or CYP3A4 such as carbamazepine, phenytoin, rifampin.

Drug Class: Cholinesterase Inhibitors				
donepezil Aricept, generics < \$30	• AD • Mixed AD/VD • LBD • PDD • VD (low evidence)	Initial: 5 mg/day PO Target: 10 mg/day PO Adjust dose after 4 wk	>10%: headache, nausea, diarrhea. <10%: vomiting, anorexia, fatigue, sleep disturbance, syncope, muscle cramps, urinary frequency. Uncommon-bradycardia, heart block prolonged QT interval.	Toxicity may be increased by inhibitors of CYP2D6 or CYP3A4, such as paroxetine, erythromycin, prednisone, grapefruit juice. Effectiveness may be reduced by inducers of CYP2D6 or CYP3A4, such as carbamazepine, phenytoin, rifampin.

Drug/Cost ^[a]	Indications	Dosage	Adverse Effects	Drug Interactions
galantamine extended-release  PAT-galantamine ER , other generics \$30–60	• Mild-moderate AD • Mild-moderate mixed AD/VD • LBD/PDD only if donepezil/rivastigmine not tolerated	Initial: 8 mg daily PO Target: 16–24 mg PO daily Adjust dose at 4-wk intervals	>10%: nausea, vomiting, diarrhea. <10%: bradycardia, syncope, dizziness, headache, sleep disturbance, fatigue, abdominal pain, weight loss, UTI , urinary incontinence, rhinitis. Uncommon: heart block, prolonged QT interval, seizures, delirium.	Toxicity may be increased by inhibitors of CYP2D6 or CYP3A4, such as paroxetine, erythromycin, prednisone, grapefruit juice. Effectiveness may be reduced by inducers of CYP2D6 or CYP3A4, such as carbamazepine, phenytoin, rifampin.
rivastigmine oral Exelon, generics \$30–60	• Mild-moderate AD • Mild-moderate mixed AD/VD • LBD • PDD	Initial: 1.5 mg BID PO Target: 6–12 mg/day PO in 2–3 divided doses [149][150] Adjust dose by 1.5 mg BID Q4 wk; take with breakfast and dinner	>10%: headache, dizziness, nausea/vomiting, diarrhea, abdominal pain, anorexia. <10%: fatigue, insomnia, syncope, dyspepsia, weight loss, UTI , rhinitis. Uncommon: heart block, prolonged QT interval, delirium, seizures.	None reported; not metabolized by cytochrome P450 system.
rivastigmine transdermal patch	• Mild-moderate AD • Mild-moderate mixed AD/VD	Initial: apply 1 Exelon-5 transdermal	>10%: headache, dizziness, nausea/vomiting,	None reported; not metabolized

Drug/Cost ^[a]	Indications	Dosage	Adverse Effects	Drug Interactions
Exelon Patch, generics \$90–120	LBD PDD	patch daily; if well tolerated, increase to Exelon-10 patch after at least 4 wk If switching from oral rivastigmine, use Exelon-5 for patients taking <3 mg BID, and Exelon-10 for patients taking 3–6 mg BID	diarrhea, abdominal pain, anorexia. <10%: fatigue, insomnia, syncope, dyspepsia, weight loss, UTI , rhinitis. Uncommon: heart block, prolonged QT interval, delirium, seizures. Health Canada warning re: accidental overdoses from failure to remove one patch before applying the next. ^[42]	by cytochrome P450 system.

Drug Class: N-methyl-D-aspartate Receptor Antagonists

memantine  Ebixa , generics \$30–60	- Adjunctive to CIs in moderate-severe AD and mixed AD/VD; LBD - Monotherapy in AD only if CIs not tolerated/contraindicated	Initial: 5 mg daily PO; increase by 5 mg daily, at weekly intervals, to 10 mg BID PO starting at wk 4	Generally well tolerated; dizziness (7%), headache (6%), confusion (6%), constipation (5%), nausea/vomiting (3%).	None reported; not affected by cytochrome P450 system. Theoretically, urinary alkalizers, such as carbonic anhydrase inhibitors, may decrease the clearance of memantine.
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Drug Class: Serotonergic Antidepressants

trazodone generics	RBD: insomnia	Initial: 25–50 mg QHS PO	Drowsiness, orthostatic	Toxicity may be increased by
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Drug/Cost ^[a]	Indications	Dosage	Adverse Effects	Drug Interactions
< \$30		Usual: insomnia: 50 mg QHS PO Other responsive behaviours: 100–200 mg/day in divided doses Maximum: 400 mg/day in divided doses	hypotension, nausea, vomiting, headache, dry mouth, priapism (rare).	inhibitors of CYP3A4, e.g., clarithromycin, erythromycin, grapefruit juice, ketoconazole. Effectiveness may be reduced by inducers of CYP3A4, such as carbamazepine, phenytoin, rifampin.

^[a] Cost of 30-day supply of target or usual dose; includes drug cost only.

 Dosage adjustment may be required in renal impairment; see [Dosage Adjustment in Renal Impairment](#).
 AD = Alzheimer dementia; CI = cholinesterase inhibitor; GI = gastrointestinal; LBD = Lewy body dementia; PDD = Parkinson disease dementia; RBD = responsive behaviours in dementia; UTI = urinary tract infection; VD = vascular dementia

Resources

Assessing driving safety in people with dementia is complicated. Helpful online resources include the [CMA Driver's Guide](#) and [CFPC recommendations](#).

Caregiver stress is common and should be anticipated. Support groups, such as those coordinated by the [Alzheimer Society of Canada](#) can help to mitigate feelings of burden.

For more information on deprescribing cholinesterase inhibitors and/or antipsychotics, see www.deprescribing.org.

Suggested Readings

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